

JAVA INTERVIEW Q/A

PART-2

? Why is the main method of Java static?

- ✓ The main reason is that the object is not required to call for a static method so, if we declare the main method as non static, we need to create an object first and then call the main() method. In order to save memory, we declare the main method as static in java.

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? Can we over ride the static method in Java?

- ✓ No, we can't over ride a static method in java.

? What are the different ways to overload a method?

- ✓ There are two different ways :-

- By changing the data type.
- By changing the number of arguments.

? What is the usage of "Super" Keywords in Java?

- ✓ The uses of Super keyword in java are :

- Super can be used to conjure an immediate parent class method.
- It is also used to indicate the immediate superclass instance variable.
- Super can be used to conjure an immediate parent class constructor.

? Can we overload main() method in java?

✓ Yes, we can overload the main() method in java by using method overloading but JVM only calls the main() method that receives string array as arguments only.

? Can java be said to be complete object oriented Programming language?

✓ Java is not a complete Object Oriented programming language because it has direct access to primitive datatypes [eg:- int, float, char, etc].

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? What is the use of JIT Compiler?

✓ JIT stands for Just In Time and it is used for improving the performance during runtime. It does the task of compiling parts of byte code having similar functionality at the same time thereby reducing the amount of compilation time for the code to run.

? What is a classloader in java?

✓ A classloader in java is a subsystem of JVM [Java Virtual Machine] dedicated to loading class files. When a program is executed, Classloader is the first to load the executable file.

? What is java string pool?

✓ A collection of strings in java's heap memory is referred to as java string pool.

In case we try to create a new string object, JVM first checks for the presence of the object in the pool. If available the same object reference is shared with the variable, else a new object is created.

? What happens when the main() isn't declared as static?

- ✓ When the main method is not declared as static then the program may be compiled correctly but ends up with a severe ambiguity and throws a runtime error: "NoSuchMethodError".

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? What is a JIT Compiler?

- ✓ JIT compiler runs after the program executed and compiles the code into a faster form, hosting the CPU's native instructing set. JIT can access dynamic runtime information, whereas a standard compiler doesn't and can be make better optimizations like intining functions that are used frequently.

? What is the Purpose of Volatile Variable?

- ✓ The value stored in a volatile variable is not read from the thread's cache memory but from the main memory. Volatile variables are primarily used during synchronization.

? What is a Wrapper class in Java?

- ✓ Wrapper class are responsible for converting the java primitives into the reference type (Objects).

A class is dedicated to every primitive datatype. They are known as wrapper class because they wrap the primitive data type into an object of that class.

It is present in java-language package.

Simple Type	Wrapper class
boolean	Boolean
char	Character
double	Double
float	Float
int	Integer
long	Long

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? What is the difference between final, finalize and finally keyword?

✓ Final :- Final is a Java keyword used to indicate that either a method can not override in a subclass, or a class can not be extended or a field can not be modified.

Finalize :- Finalize is a method that gets called on an instance of an object when it is garbage collected.

Finally :- Finally is a Java keyword used in exception handling to indicate a block of code that should always be run whether an exception is thrown or not.

? How does Garbage collection prevent a Java application from getting out of memory?

✓ Garbage collection simply cleans up unused memory when an object goes out of scope and is no longer needed.

However an application could create a huge number of large objects that causes an **OutOfMemoryError**.

? What is the difference between a **ClassNotFoundException** and **NoClassDefFoundError**?

✓ **ClassNotFoundException**:- A **ClassNotFoundException** means the class file for a requested class is not on the class path of the application.

NoClassDefFoundError:- A **NoClassDefFoundError** means that the class file existed at runtime, but for some reason that class could not be turned into class definition.

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? What is the contract between **equals** and **hashCode** of an object?

✓ The only obligation is that for any objects **o1** and **o2** if,
o1.equals(o2) is true then

o1.hashCode() == o2.hashCode() is true
also

for any **o1, o2** of some class **C**, where none of **o1** and **o2** are null

, then it can happen that

o1.hashCode() == o2.hashCode() is true but **o1.equals(o2)** is false.

? What is the difference between Hash Map and Concurrent Hashmap?

- ✓ Concurrent Hash Map is thread - safe that is the code can be accessed by single thread at a time while Hashmap is not thread - safe.
ConcurrentHashMap does not allow NULL keys while HashMap allows it.